education

Department:
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11

NATURAL SCIENCE
MARKING GUIDELINES

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MATHEMATICS N5
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WISKUNDE N5

Question / Vraag 1

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1.1 $\lim_{x \to 0} y = \lim_{x \to 0} \frac{x \cdot e^x}{e^x - 1}$

1.1.1 $\lim_{x \to 0} y = \lim_{x \to 0} \frac{x \cdot e^x}{e^x - 1} = \frac{0}{0}$ .... indeterminate / onbepaald

$= \lim_{x \to 0} \left[\frac{1 \cdot e^x + x \cdot e^x}{e^x} \right]$

$= \lim_{x \to 0} (1 + x)$

$\therefore \ln y = 1$ ✓ (3)

1.1.2 $y = e^1 = 2,718$ ✓ (1)

1.2 For discontinuity / Vir diskontinuïteit: $2x^2 - x - 3 = 0$ (2)

$\therefore (2x - 3)(x + 1) = 0$

$\therefore x = \frac{3}{2}$ ✓ or $x = -3$ ✓

[6]

Question / Vraag 2

2.1 $f(x) = -\sqrt{2x} = -\sqrt{2} \cdot x^{\frac{1}{2}}$

2.1.1 $f(x+h) = -\sqrt{2}\,(x+h)^{\frac{1}{2}}$

$= -\sqrt{2} \left(x^{\frac{1}{2}} + \frac{\frac{1}{2} x^{-\frac{1}{2}} h^1}{1!} + \frac{\frac{1}{2} \cdot -\frac{1}{2} x^{-\frac{3}{2}} h^2}{2!} + \ldots \right)$

$f(x+h) = -\sqrt{2} \cdot x^{\frac{1}{2}} - \frac{\sqrt{2}}{2} x^{-\frac{1}{2}} h^1 + \frac{\sqrt{2}}{8} x^{-\frac{3}{2}} h^2 + \ldots$ (2)

2.1.2 $f(x+h) - f(x) = -\sqrt{2}.x^{\frac{1}{2}} - \frac{\sqrt{2}}{2}x^{-\frac{1}{2}}h^1 + \frac{\sqrt{2}}{8}x^{-\frac{3}{2}}h^2 + \ldots - (-\sqrt{2}.x^{\frac{1}{2}})$

$f(x+h) - f(x) = -\frac{\sqrt{2}}{2}x^{-\frac{1}{2}}h^1 + \frac{\sqrt{2}}{8}x^{-\frac{3}{2}}h^2 + \ldots$ ✓ ✓ ✓ 1

2.1.3 $\frac{f(x+h) - f(x)}{h} = \frac{-\frac{\sqrt{2}}{2}x^{-\frac{1}{2}}h^1 + \frac{\sqrt{2}}{8}x^{-\frac{3}{2}}h^2 + \ldots}{h}$

$\frac{f(x+h) - f(x)}{h} = -\frac{\sqrt{2}}{2}x^{-\frac{1}{2}} + \frac{\sqrt{2}}{8}x^{-\frac{3}{2}}h + \ldots$ ✓ ✓ ✓ 1

2.1.4 $\lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = -\frac{\sqrt{2}}{2}x^{-\frac{1}{2}}$ ✓ (1)

2.2 $y = \text{arc cosec } x$

$\text{cosec } y = x$

$-\text{cosec } y.\cot y.\frac{dy}{dx} = 1$ ✓

$\frac{dy}{dx} = \frac{-1}{\text{cosec } y.\cot y}$ ✓ where $\cot y = \sqrt{\text{cosec}^2 y - 1}$

$\frac{dy}{dx} = \frac{-1}{\text{cosec } y.\sqrt{\text{cosec}^2 y - 1}}$ ✓ (3)

$\therefore \frac{dy}{dx} = \frac{-1}{x\sqrt{x^2-1}}$

2.3.1 $\frac{dy}{dx} = \frac{1}{(e^{2x}.x^5)}\left[2e^{2x}.x^5 + 5x^4.e^{2x}\right]$ ✓ ✓ ✓ (3)

2.3.2 $\frac{dy}{dx} = \frac{1}{2}(\ln x)^{-\frac{1}{2}}.\frac{1}{x} - \frac{1}{\tan x \ln 7.\sec^2 x}$ ✓ ✓ ✓ ✓ (3)

2.3.3 $\frac{dy}{dx} = e^{\left(\frac{\cos x}{\ln x}\right)}.\left[\frac{(-\sin x)(\ln x) - \left(\frac{1}{x}\right)(\cos x)}{(\ln x)^2}\right]$ ✓ ✓ ✓ (3)

2.4 $y = (\cos x)^{\ln(\sin x)}$

$\ln y = \ln(\cos x)^{\ln(\sin x)}$

$\ln y = \ln(\sin x).\ln(\cos x)$

$\frac{1}{y}.\frac{dy}{dx} = \left(\frac{1}{\sin x}.\cos x\right)\ln(\cos x) + \left(\frac{1}{\cos x}.-\sin x\right)\ln(\sin x)$

$\frac{1}{y}.\frac{dy}{dx} = \cot x.\ln(\cos x) - \tan x.\ln(\sin x)$

$\frac{dy}{dx} = y\left[\cot x.\ln(\cos x) - \tan x.\ln(\sin x)\right]$ (4)

$\frac{dy}{dx} = (\cos x)^{\ln(\sin x)}\left[\cot x.\ln(\cos x) - \tan x.\ln(\sin x)\right]$

2.5 $\ln(e^{x^2}) - \cos 2y = y^2$

$x^2 - \cos 2y = y^2$

$2x + 2\sin 2y.\frac{dy}{dx} = 2y.\frac{dy}{dx}$

$\frac{dy}{dx}(2\sin 2y - 2y) = -2x$

$\frac{dy}{dx} = \frac{-2x}{2\sin 2y - 2y}$ (4)

$\frac{dy}{dx} = \frac{-2(2)}{2\sin 2\left(\frac{\pi}{12}\right) - 2\left(\frac{\pi}{12}\right)}$

$\frac{dy}{dx} = -8,346$

[25]

Question / Vraag 3

3.1 $S = 2t^3 - 12t^2 + 9t$

3.1.1 $V = \frac{dS}{dt} = 6t^2 - 24t + 9$ ✓ ✓ (1)

3.1.2 $a = \frac{dV}{dt} = 12t - 24$ ✓ (1)

3.1.3 At rest, $V = 0\ m.s^{-1}$

$\therefore 6t^2 - 24t + 9 = 0$ ✓

$2t^2 - 8t + 3 = 0$ (3)

$t = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(2)(3)}}{2(2)}$

$t = 3,581\ s$ ✓ and/en $t = 0,419\ s$ ✓

3.2 $xy = 25 \quad \therefore y = \frac{25}{x}$ ✓

$S = x + y$

$S = x + \frac{25}{x}$ ✓

$\frac{dS}{dx} = 1 - \frac{25}{x^2} = 0$ ✓ (5)

$x = -5$ ✓ and/en $x \neq 5$

$\therefore y = \frac{25}{-5}$

$y = -5$ ✓

3.3 $V = \pi r^2 h$

$\frac{dV}{dt} = \pi r^2 . \frac{dh}{dt}$ ✓

$5400 = \pi (300)^2 . \frac{dh}{dt}$ ✓ ✓ (3)

$\therefore \frac{dh}{dt} = \frac{3}{50\pi}$ cm.s^{-1} ✓

3.4 $f(x) = \frac{1}{2}x^3 - 3x^2 + 5x - 2$

3.4.1 $f'(x) = \frac{3}{2}x^2 - 6x + 5 = 0$

$3x^2 - 12x + 10 = 0$

$x = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(3)(10)}}{2(3)}$

$x = 1,184$ and/en $x = 2,817$ ✓

$\therefore y = 0,544$ ✓ $\therefore y = -0,544$ ✓ (2)

3.4.2

x	0	1	2	3	4
$f(x)$	-2	0,5	0	-0,5	2

✓ (1)

3.4.3 Let/Laat $x_1 = 0,6$

$f(0,6) = 0,028$ ✓

$f'(0,6) = 1,940$ ✓

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ (3)

$= 0,6 - \left(\frac{0,028}{1,940}\right)$ ✓

$x_2 = 0,586$ ✓

✓ ✓

[11]

Question / Vraag 4

4.1.1 $\int e^{\frac{1}{2}x}.\cos x.dx = e^{\frac{1}{2}x}.\sin x - \int \frac{1}{2}e^{\frac{1}{2}x}.\sin x.dx$

$= e^{\frac{1}{2}x}.\sin x - \frac{1}{2}\left(e^{\frac{1}{2}x}.-\cos x - \int \frac{1}{2}e^{\frac{1}{2}x}.-\cos x.dx\right)$

$\int e^{\frac{1}{2}x}.\cos x.dx = e^{\frac{1}{2}x}.\sin x + \frac{1}{2}e^{\frac{1}{2}x}.\cos x - \frac{1}{4}\int e^{\frac{1}{2}x}.\cos x.dx$

$\frac{5}{4}\int e^{\frac{1}{2}x}.\cos x.dx = e^{\frac{1}{2}x}.\sin x + \frac{1}{2}e^{\frac{1}{2}x}.\cos x$ (4)

$\int e^{\frac{1}{2}x}.\cos x.dx = \frac{4}{5}e^{\frac{1}{2}x}.\sin x + \frac{2}{5}e^{\frac{1}{2}x}.\cos x + C$

4.1.2 $\int \frac{\tan x}{\ln(\sec x)}.dx$ $u = \ln(\sec x)$

$= \int \frac{1}{\ln(\sec x)}.\tan x.dx$ $du = \frac{1}{\sec x}.\sec x.\tan x.dx$

$= \int \frac{1}{u}.du$ $du = \tan x.dx$

$= \ln u + C$

$= \ln(\ln(\sec x)) + C$ (3)

4.1.3 $\int \sqrt{(\ln x + e^{-x})^3}.\left(\frac{1}{x} - e^{-x}\right).dx$ $u = \ln x + e^{-x}$

$= \int (\ln x + e^{-x})^{\frac{3}{2}}.\left(\frac{1}{x} - e^{-x}\right).dx$ $du = \left(\frac{1}{x} - e^{-x}\right).dx$

$= \int u^{\frac{3}{2}}.du$

$= \frac{2}{5}u^{\frac{5}{2}} + C$

$= \frac{2}{5}\sqrt{(\ln x + e^{-x})^5} + C$ (3)

4.1.4 $\int \cos^3 x \, dx$ $\quad u = \sin x$

$= \int \cos^2 x \cdot \cos x \cdot dx$ $\quad du = \cos x \cdot dx$

$= \int (1 - \sin^2 x) \cdot \cos x \cdot dx$

$= \int (1 - u^2) \cdot du$

$= u - \frac{1}{3}u^3 + C$

$= \sin x - \frac{1}{3}\sin^3 x + C$ (3)

4.1.5 $\int \frac{\sec^2 x - \tan^2 x}{\sec x + \tan x} \cdot dx$

$= \int \frac{(\sec x + \tan x)(\sec x - \tan x)}{\sec x + \tan x} \cdot dx$

$= \int (\sec x - \tan x) \cdot dx$

$= \ln(\sec x + \tan x) - \ln(\sec x) + C$ (3)

4.1.6 $\int \frac{1}{4 + 16x^2} \cdot dx$

$= \frac{1}{16} \int \frac{1}{\frac{1}{4} + x^2} \cdot dx$

$= \frac{1}{16} \cdot \frac{1}{\frac{1}{2}} \tan^{-1}\left(\frac{x}{\frac{1}{2}}\right) + C$

$= \frac{1}{8} \tan^{-1}(2x) + C$ (3)

4.2 $\frac{5 + x}{x^2 - 2x} = \frac{A}{x} + \frac{B}{x - 2}$

$5 + x = A(x - 2) + Bx$

let/laat $x = 0 \quad \therefore A = -\frac{5}{2}$ ✓

$x = 2 \quad \therefore B = \frac{7}{2}$ ✓

$$\int \frac{5+x}{x^2-2x}\,.\,dx = -\frac{5}{2}\int \frac{1}{x}\,.\,dx + \frac{7}{2}\int \frac{1}{(x-2)}\,.\,dx$$

$$= -\frac{5}{2}\ln x + \frac{7}{2}\ln(x-2) + C \quad (5)$$

[24]

Question / Vraag 5

5.1 $\int_0^{\infty} e^{-st}\,.\,f(t)\,.\,dt$ if $f(t) = -\frac{1}{2}$

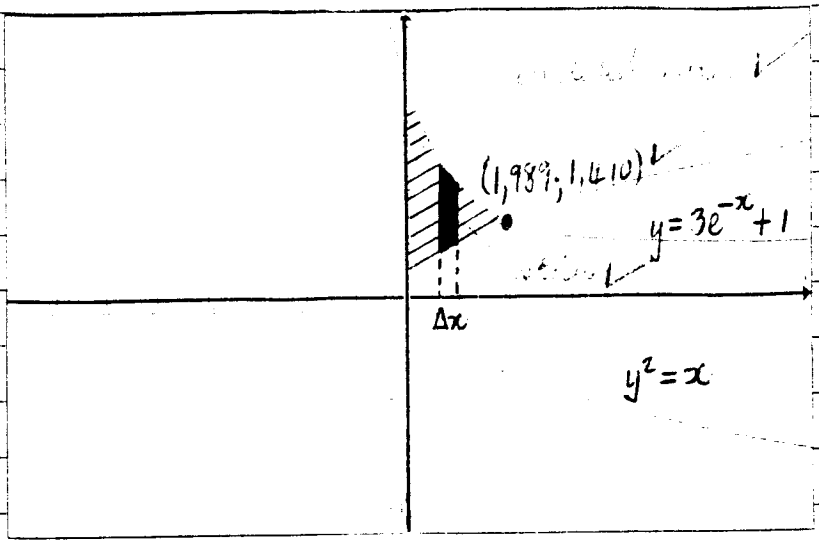
$$= \int_0^{\infty} e^{-st}\,.\,-\frac{1}{2}\,.\,dt$$

$$= -\frac{1}{2}\left[-\frac{1}{s}e^{-st}\right]_0^{\infty}$$

$$= -\frac{1}{2}\left[(0) - \left(-\frac{1}{s}\right)\right]$$

$$= -\frac{1}{2s} \quad (4)$$

5.2.1



(3)

5.2.2 $\delta A = (y_2 - y_1)\,\delta x$

$\delta A = \left((3e^{-x} + 1) - \sqrt{x}\right).\,\delta x$

$A = \int_0^{1,989} (3e^{-x} + 1 - \sqrt{x}).\,dx$

$= \left[-3e^{-x} + x - \frac{2}{3}x^{\frac{3}{2}}\right]_0^{1,989}$

(3)

$A = 2,708 \text{ units}^2$

5.2.3 $\delta V = \pi\,(y_2^2 - y_1^2).\,\delta x$

$\delta V = \pi\left((3e^{-x} + 1)^2 - (\sqrt{x})^2\right).\,\delta x$

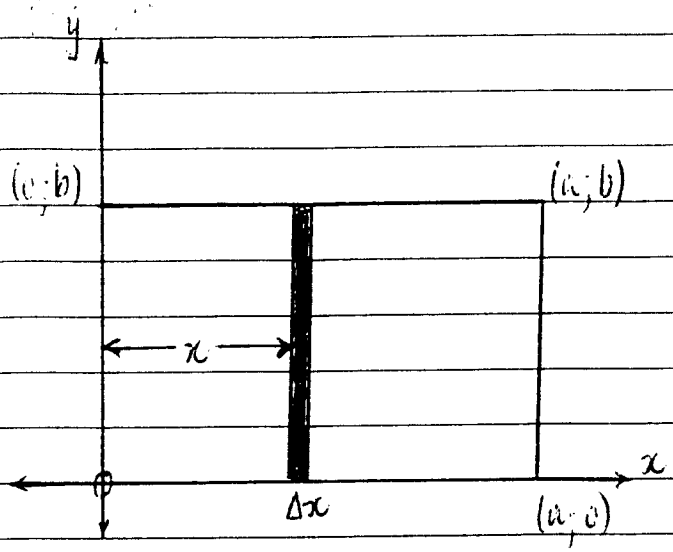
$V = \pi \int_0^{1,989} (9e^{-2x} + 6e^{-x} + 1 - x).\,dx$

$= \pi\left[-\frac{9}{2}e^{-2x} - 6e^{-x} + x - \frac{1}{2}x^2\right]_0^{1,989}$

(4)

$V = 30,177 \text{ units}^3$

5.3



$\therefore \Delta m = b\Delta x \times \frac{m}{ab}$

$= \frac{m}{a} . \Delta x$ ✓

$\therefore \Delta I = \Delta m \times x^2$

$= \frac{m}{a} . \Delta x \times x^2$ ✓

$\therefore I = \frac{m}{a} \int_0^a x^2 . dx$

$= \frac{m}{a} \left[\frac{x^3}{3}\right]_0^a$ ✓

$I = \frac{1}{3} m a^2$ kg. units^2 ✓

(4)

[13]

Question / Vraag 6

6.1 $\frac{dy}{dx} = \frac{1}{x} - \sin x$

$dy = \left(\frac{1}{x} - \sin x\right) . dx$

$\int 1 . dy = \int \left(\frac{1}{x} - \sin x\right) . dx$ ✓ ✓

$y = \ln x + \cos x + C$ ✓ ✓

(2)

6.2 $\sec^2 y \cdot \frac{dy}{dx} = (1 + \tan y)$

$\frac{\sec^2 y}{(1 + \tan y)} \cdot dy = 1 \cdot dx$

$\int \frac{\sec^2 y}{(1 + \tan y)} \cdot dy = \int 1 \cdot dx$

$\ln(1 + \tan y) = x + C$ (3)

6.3 $-\frac{d^2y}{dx^2} = x^2 - 1$

$\frac{d^2y}{dx^2} = -x^2 + 1$

$\frac{dy}{dx} = -\frac{1}{3}x^3 + x + C_1$

$y = -\frac{1}{12}x^4 + \frac{1}{2}x^2 + C_1 x + C_2$ (5)

[8]

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